Introduction to Functions

Fiona Young

Why functions?

EXAMPLE How does your teacher know what grade to give you on a test? A *function* relates:

90 out of 100 points $\longrightarrow$ grade of A^{-}

EXAMPLE You press a button on the soda machine when you're thirsty. A *function* relates:

The "Sprite" button $\longrightarrow$ Actual Sprite

EXAMPLE How do you tell your computer what color you want your website background to be? A *function* relates:

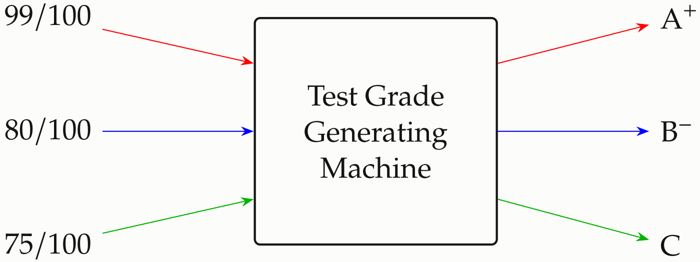
#FF5733 $\longrightarrow$

EXAMPLE You want a $3 shirt but you only have quarters. A *function* relates:

$3 $\longrightarrow$ 12 quarters

So what exactly is a function?

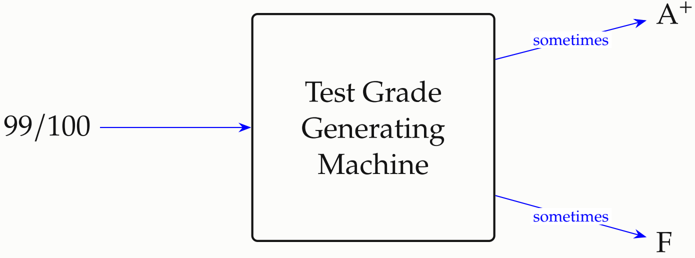
A function is like a machine that takes inputs and turns them into outputs using a predetermined recipe.



So what exactly is a function?

A function must be *well-defined*, which means that for every unique input, there must be a unique output.

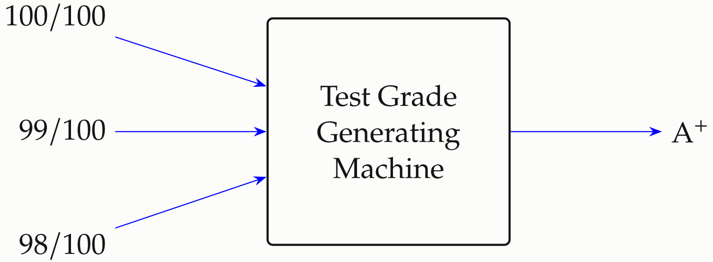
Why? Consider the following scenario: the Test Grade Generating Machine sometimes outputs A^+ for a score of 99/100, but other times it outputs F.



That would be inconsistent, and pretty unfair!

So what exactly is a function?

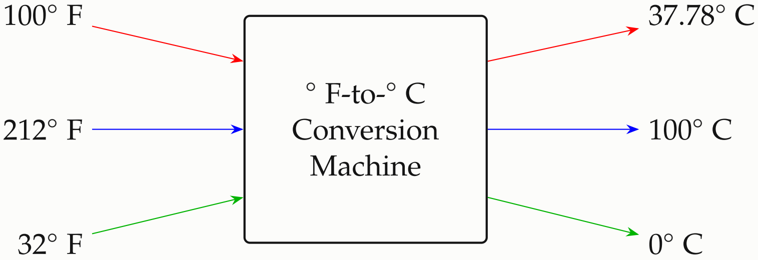
However, a function is allowed to send different inputs to the same outputs. (Pause and think: how is this different from being *well-defined*?)



Sometimes the information we input is more *detailed* than the information we need outputted. A function is allowed to get rid of some of that extra detail. If we simply need the letter grade, then the exact numerical score (98, 99, vs. 100) doesn't matter.

One-to-one functions

Some functions do *not* get rid of extra details. Consider the Fahrenheit-to-Celsius Conversion function:



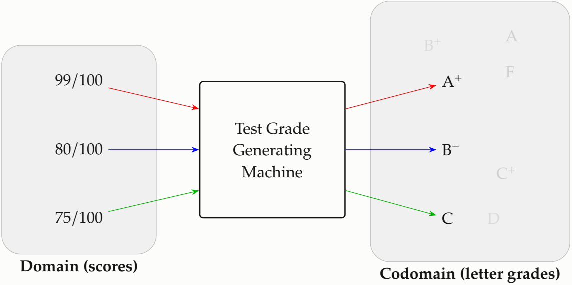
Can you think of two *different* degrees in Fahrenheit that convert to the *same* degree in Celsius? (It is impossible.)

A function for which each unique output comes from a unique input is called *one-to-one* (or *injective*).

The domain and codomain of a function

The *domain* of a function is like the category of the inputs, and the *codomain* is the category of the outputs.

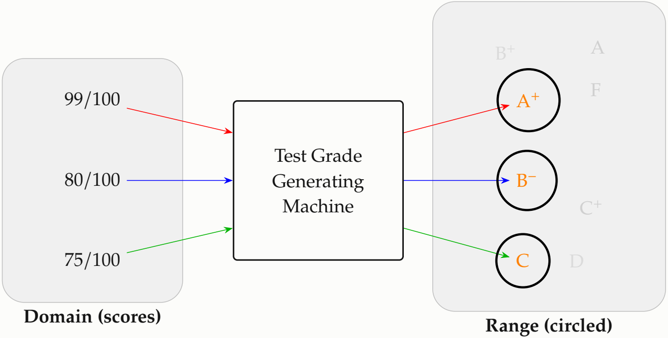
Example For our Test Grade Generating Machine, the domain is *scores* and the codomain is *letter grades*. Notice how there are values in the codomain that do not have arrows attached to them.



The range of a function

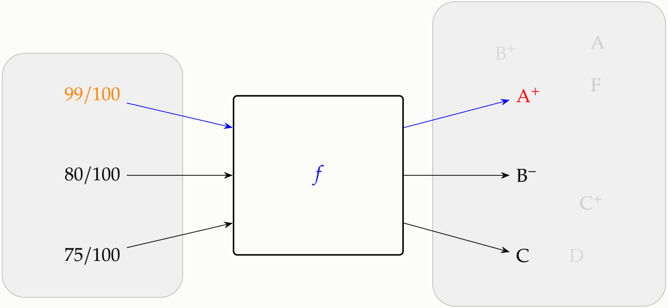
The *range* is the set of outputs that actually occur. Do not confuse it with the *codomain*.

EXAMPLE For our Test Grade Generating Machine, the range is $\{A^+, B^-, C\}$ while the codomain is all letter grades (from A+ all the way down to F).



How mathematicians write functions

Instead of saying "the Test Grade Generating Machine" mathematicians simply give the function a name, like f.



To convey that the machine takes 99/100 from the domain and produces A$^+$ in the codomain, we write $\boxed{f(99/100) = \mathrm{A}^+}$. More on this in the next lesson!